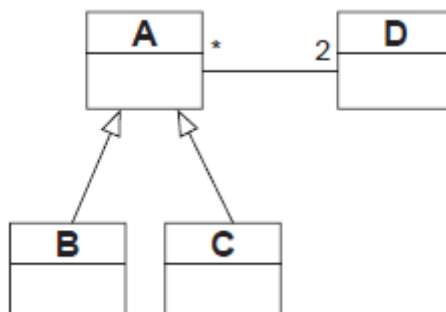


الإجابات على "كراس الإجابة"

Treat the following questions:**I- Choose the correct answer: (1pt/correct answer)****1- In following class diagram, which of following statements is true?**

- a- One object of B must be associated with many objects of D.
- b- There are no direct instances of A.
- c- Two objects of D are associated with * objects of A.
- d- One object of D may be associated with multiple objects of C.

2- The fact that the same operation may apply to two or more classes is called what?

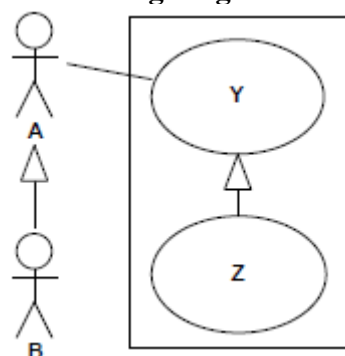
- a- Inheritance
- b- Polymorphism
- c- Encapsulation
- d- Multiple classification

3- In a class diagram, a black diamond represents:

- a- An aggregation
- b- A composition
- c- A generalization
- d- None of the above

4- In a Use Case diagram, we split the future software into packages for:

- a- Make work easier and see clearly.
- b- Analyze features separately.
- c- Resolve the different needs of actors.
- d- All of the above.

5- Which of the following statements about the following diagram are true?

- a- Only A can execute Z.
- b- A and B execute Z together.
- c- A and B can execute Y separately.
- d- A and B execute Y together.

6- How do you model the following situation: “An order is made with exactly one waiter, one waiter handles multiple orders.”?

- a-
- b-
- c-
- d-

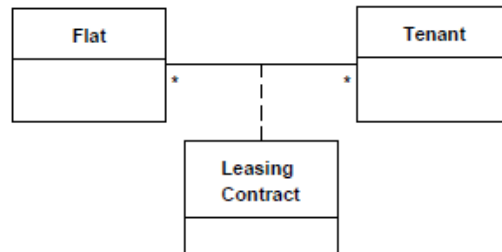
7- An <<include>> relationship means, that:

- a- The behavior of B always has to be inserted into the behavior of A.
- b- The behavior of B might or might not be inserted into the behavior of A.
- c- The behavior of A always has to be inserted into the behavior of B.
- d- The behavior of A might or might not be inserted into the behavior of B.

8- An abstract class is which of the following?

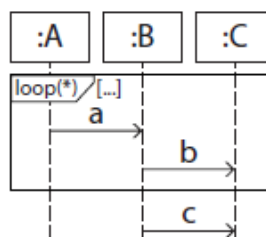
- a- A class that has direct instances, but whose descendants may have direct instances.
- b- A class that has no direct instances, but whose descendants may have direct instances.
- c- A class that has direct instances, but whose descendants may not have direct instances.
- d- A class that has no direct instances, but whose descendants may not have direct instances

9- In the following class diagram. Which of the following statements is true?



- a- One tenant can rent multiple flats without leasing contracts.
- b- One flat can be rented by multiple tenants without leasing contracts.
- c- One tenant rents exactly one flat.
- d- Many tenants can rent many flats with different leasing contracts.

10- In the following sequence diagram, which traces are possible?



- a- $b \rightarrow a \rightarrow c$
- b- $a \rightarrow b \rightarrow b \rightarrow c$
- c- $a \rightarrow b \rightarrow c \rightarrow a \rightarrow b$
- d- $a \rightarrow b \rightarrow a \rightarrow b \rightarrow a \rightarrow b \rightarrow c$

II- Open Questions: (2pts/ question)

1- In a Class Diagram there are: Dependencies and Generalization. What is the difference between them?

2- Model this situation using a sequence diagram:

To place an order, the customer provides his phone number (for identification) and the contents of his order in form of a list. The system starts by verifying that the customer exists in the customer base and keeps the corresponding object. The system then checks that the order is feasible (can be manufactured), then creates the order in the system and reports the order number and the price to the customer. Then the system assigns the customer to the order. It keeps the created order object, then returns the order number and the price to the customer.

3- In a use case diagram, what is the difference between an include relation and an exclude relation?

4- How do you model the following situation with a Class diagram: A doctor treats several patients; a patient can be treated by more than one doctor. To input which doctor treated which patient, the date and time of each treatment is recorded. It should be possible for a doctor to treat the same patient more than once.

5- Draw the cash dispenser use case diagram

- a- A customer can make a transfer, withdraw money and view their account.
- b- Employees can perform the same operations.
- c- They are also responsible for carrying out maintenance operations.
- d- Anyone who interacts with the system must authenticate.